 SS(Sem-II) — Phy (7)

2023

Time : 3 hours

Full Marks : 70

Pass Marks : 32

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer any **five** questions, in which Q. No. 1 is compulsory.*

1. Answer any **four** questions of the following :

$3\frac{1}{2} \times 4 = 14$

(a) Write down the physical meaning of all quantum numbers used to solve the problem of Hydrogen atom.

(b) What is the significance of Stern-Gerlach experiment ?

(c) State Lande's interval rule.

(d) Write down the applications of NMR.

(e) Define spontaneous and stimulated emission of radiation.

(f) Calculate possible orientation (space quantization) of the orbital angular momentum vector $\ell = 1$ with respect to the magnetic field direction.

2. Discuss the fine structure spectrum of the Hydrogen atom by considering the spin-orbit interaction only. 14

3. Explain the L-S and J-J coupling schemes for two valance electron atoms. 7+7 = 14

4. Define Normal and Anomalous Zeeman effects. Discuss the theory of Anomalous Zeeman effect. 4 + 10 = 14

5. Define pure rotational and vibration-rotational spectra. Discuss the theory of pure rotational spectra by assuming the molecule as a rigid rotator. 4 + 10 = 14

6. What is Raman Effect ? Explain the quantum mechanical theory of Raman effect. 4 + 10 = 14

7. What do you mean by Laser Rate Equation ? Deduce the condition of population inversion in three level laser system using this equation.

$4 + 10 = 14$

8. Describe with a neat diagram, the construction and working of Ruby Laser. 14

9. Write short notes on any **two** of the following :

$7 \times 2 = 14$

(a) Electronic Spectra

(b) ESR

(c) Stark Effect

(d) Paschen-Back Effect